Báo Cáo thực hành lab04
Lập trình hướng đối tượng
Trần mạnh hùng – 20220068

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2. Additional requirements of AIMS

3. Creating the **Book** class

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;


import java.util.*;
public class Book extends Media {
    private List<String> authors = new ArrayList<String>();

    public Book(int id, String title, String category, float cost) {
        super(id, title, category, cost);
    }

    public Book(int id, String title, String category, float cost, List<String> authors) {
        super(id, title, category, cost);
        this.authors = authors;
    }

    public List<String> getAuthors() {
        return authors;
    }

    public void setAuthors(List<String> authors) {
        this.authors = authors;
    }

    public void addAuthor(String authorName) {
        if (!authors.contains(authorName)) {
            authors.add(authorName);
            System.out.println(authorName+" has been added to the list.");
        }
        System.out.println(authorName+" is already in the list.");
    }

    public void removeAuthor(String authorName) {
        if (!authors.contains(authorName)) {
            System.out.println(authorName+" is not in the list.");
        }
        else {
```

```java
        else {
            authors.remove(authorName);
            System.out.println(authorName+" has been removed from the list.");
        }
    }

    public String toString() {
        return "ID - "+ getId()+ ". Book - " + getTitle() + " - " + getCategory() + " - " + getAuthors() + " - $" + getCost();
    }
}
```

4. Creating the abstract **Media** class

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;

import java.util.ArrayList;
import java.util.Collections;
import java.util.Comparator;

public abstract class Media {
    private int id;
    private String title;
    private String category;
    private float cost;

    public static final Comparator<Media> COMPARE_BY_TITLE_COST =
            new MediaComparatorByTitleCost();
    public static final Comparator<Media> COMPARE_BY_COST_TITLE =
            new MediaComparatorByCostTitle();

    public Media(int id, String title, String category, float cost) {
        super();
        this.id = id;
        this.title = title;
        this.category = category;
        this.cost = cost;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getTitle() {
        return title;
    }

```

```java
    Run | Debug
    public static void main(String[] args) {
        ArrayList<Media> mediae = new ArrayList<>();

        CompactDisc cd = new CompactDisc(id:1,title:"Soledad", category:"Ballad", cost:12.5f);
        DigitalVideoDisc dvd = new DigitalVideoDisc(id:3,title:"Final Fantasy X", category:"Fantasy", cost:222.22f );
        Book book = new Book(id:2,title:"Operating System Concepts", category:"ICT", cost:30f);

        // Add some media objects to the list
        mediae.add(cd);
        mediae.add(dvd);
        mediae.add(book);

        Collections.sort(mediae, Media.COMPARE_BY_TITLE_COST);

        for (Media media : mediae) {
            System.out.println(media.toString());
        }
    }
}
```

```java

    public void setTitle(String title) {
        this.title = title;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public float getCost() {
        return cost;
    }

    public void setCost(float cost) {
        this.cost = cost;
    }

    public boolean equals(Object obj) {
        if (obj instanceof Media) {
            Media media = (Media) obj;
            if (this.id == media.id) {
                return true;
            }
        }
        return false;
    }

    Run | Debug
    public static void main(String[] args) {
        ArrayList<Media> mediae = new ArrayList<>();
```

Modify Book and DigitalVideoDisc classes to extend Media

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;


import java.util.*;
public class Book extends Media {
    private List<String> authors = new ArrayList<String>();

    public Book(int id, String title, String category, float cost) {
        super(id, title, category, cost);
    }

    public Book(int id, String title, String category, float cost, List<String> authors) {
        super(id, title, category, cost);
        this.authors = authors;
    }

    public List<String> getAuthors() {
        return authors;
    }

    public void setAuthors(List<String> authors) {
        this.authors = authors;
    }
```

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;

public class DigitalVideoDisc extends Disc implements Playable {
    public DigitalVideoDisc(int id, String title, String category, float cost) {
        super(id, title, category, cost);
    }

    public void play() {
        System.out.println("Playing DVD: " + this.getTitle());
        System.out.println("DVD length: " + this.getLength());
    }

    //toString method
    public String toString() {
        return "ID - " +getId()+ ". DVD - " + getTitle() + " - " + getCategory() + " - " + getDirector() + " - " + getLen
    }
}

```

5. Creating the **CompactDisc** class

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;

public class Disc extends Media {
    private int length;
    private String director;

    public Disc(int id, String title, String category, float cost) {
        super(id,title, category, cost);
    }

    public Disc(int id, String title, String category, float cost, int length, String director) {
        super(id, title, category, cost);
        this.length = length;
        this.director = director;
    }

    public int getLength() {
        return length;
    }
    public void setLength(int length) {
        this.length = length;
    }
    public String getDirector() {
        return director;
    }
    public void setDirector(String director) {
        this.director = director;
    }

}
```

Create the Track class

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;

public class Track extends CompactDisc implements Playable {
    private String titleTrack;
    private int length;

    public Track(int id, String title, String category, float cost, String titleTrack, int length) {
        super(id, title, category, cost);
        this.titleTrack = titleTrack;
        this.length = length;
    }

    public void play() {
        System.out.println("Playing DVD: " + this.getTitle());
        System.out.println("DVD length: " + this.getLength());
    }

    public String getTitle() {
        return titleTrack;
    }
    public void setTitle(String title) {
        this.titleTrack = title;
    }
    public int getLength() {
        return length;
    }
    public void setLength(int length) {
        this.length = length;
    }

//<<<<<<< topic/override-equals-method
    public boolean equals(Object obj) {
        if (obj instanceof Track) {
            Track track = (Track) obj;
            if (this.titleTrack.equals(track.titleTrack) && this.length == track.length) {
                return true;
            }
```

Create the CompactDisc class

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;

import java.util.ArrayList;

public class CompactDisc extends Disc implements Playable {
    private String artist;
    private ArrayList<Track> tracks = new ArrayList<Track>();

    public CompactDisc(int id, String title, String category, float cost) {
        super(id, title, category, cost);
    }

    public CompactDisc(int id, String title, String category, float cost, String artist, ArrayList<Track> tracks) {
        super(id, title, category, cost);
        this.artist = artist;
        this.tracks = tracks;
    }

    public void play() {
        System.out.println("Playing CD: " + this.getTitle());
        System.out.println("CD length: " + this.getLength());
        for (Track track : tracks) {
            track.play();
        }
    }

    public void addTrack(Track track) {
        if(!tracks.contains(track)) {
            tracks.add(track);
            System.out.println(track+" has been added");
        }
        System.out.println(track+" is already in the track list");
    }

    public void removeTrack(Track track) {
        if(!tracks.contains(track)) {
            System.out.println(track+" is not in the track list");
```

6. Create the Playable interface and Modify the DigitalVideoDisc class

lab04 > Aimsproject > hust > soict > hedspi > aims > media > Playable.java > Playable

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;
public interface Playable {
    public void play();
}
```

7. Update the **Cart** class to work with **Media**

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.cart;

import java.util.ArrayList;

import lab04.Aimsproject.hust.soict.hedspi.aims.media.Media;

public class Cart {
    private final int MAX_NUMBER_ORDERED = 20;
    private ArrayList<Media> itemsOrdered = new ArrayList<Media>(MAX_NUMBER_ORDERED);

    public void addMedia(Media media) {
        if (itemsOrdered.size()<MAX_NUMBER_ORDERED) {
            itemsOrdered.add(media);
            System.out.println(media.getTitle()+" has been added to the cart.");
        }
        else System.out.println(x:"The cart is full.");
    }

    public void removeMedia(Media media) {
        boolean found = false;
        for (Media item : itemsOrdered) {
            if (item.equals(media)) {
                itemsOrdered.remove(item);
                System.out.println(media.getTitle() + " has been removed from the cart.");
                found = true;
                break;
            }
        }
        if (!found) {
            System.out.println(media.getTitle() + " is not in the cart.");
        }
    }

    public ArrayList<Media> getItemsInCart(){
        return itemsOrdered;
    }

```

```java
    //Get total cost
    public float totalCost() {
        float total = 0.0f;
        for (int i = 0; i < itemsOrdered.size(); i++) {
            total += itemsOrdered.get(i).getCost();
        }
        return total;
    }

    //Print cart method
    public void printCart() {
        System.out.println(x:"***********************CART************************");
        System.out.println(x:"Ordered Items:");
        for (int i = 0 ; i < itemsOrdered.size(); i++) {
            System.out.println((i+1) + itemsOrdered.get(i).toString());
        }
        System.out.println("Total cost: $" + totalCost());
        System.out.println(x:"****************************************************");
    }

    //Search for DVDs in the cart by ID and display them
    //Notify to user if no match is found
    public void searchID(int id){
        boolean found = false;
        for (int i = 0; i < itemsOrdered.size(); i++) {
            if (itemsOrdered.get(i).getId() == id) {
                found = true;
                System.out.println(x:"DVD found: ");
                System.out.println(itemsOrdered.get(i).toString());
                break;
            }
```

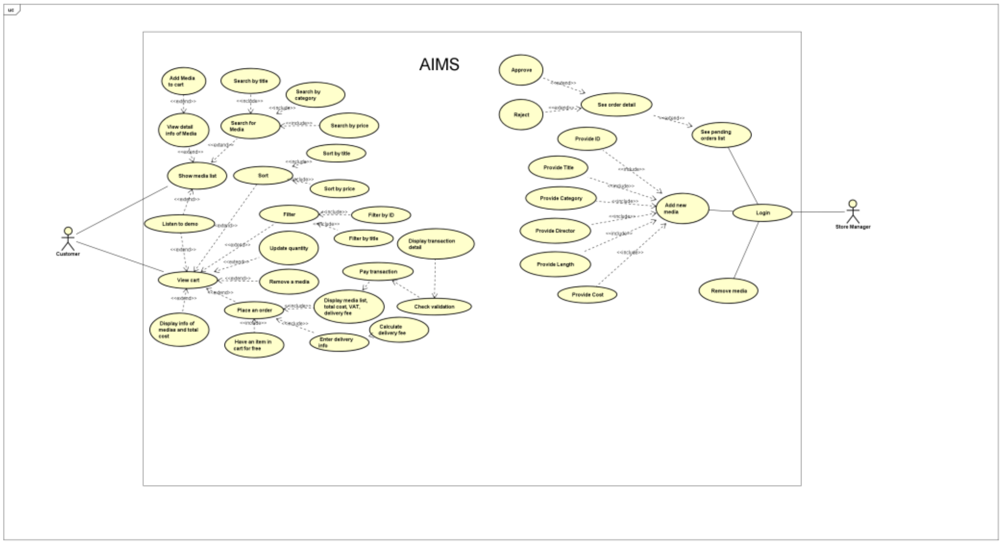
```java
        }
        if (!found) System.out.println(x:"No match is found!");
    }

    //Search for DVDs in the cart by title and display them
    //Notify to user if no match is found
    public void searchTitle(String title){
        boolean found = false;
        for (int i = 0; i < itemsOrdered.size(); i++) {
            if (itemsOrdered.get(i).getTitle().equals(title)) {
                found = true;
                System.out.println(x:"DVD found: ");
                System.out.println(itemsOrdered.get(i).toString());
                break;
            }
        }
        if (!found) System.out.println(x:"No match is found!");
    }
}
```

8. Update the **Store** class to work with **Media**

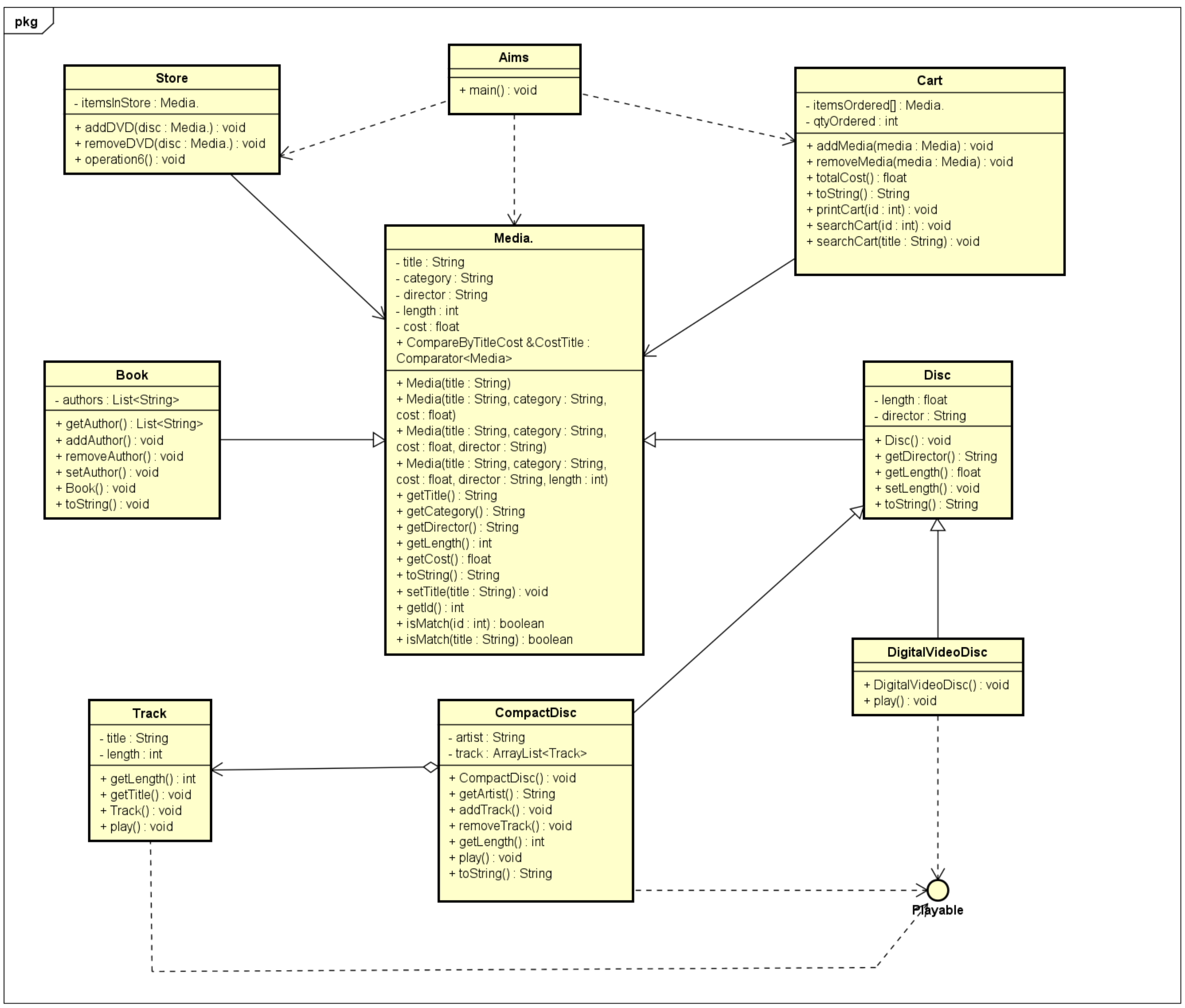
```java
package lab04.Aimsproject.hust.soict.hedspi.aims.store;

import java.util.ArrayList;

import lab04.Aimsproject.hust.soict.hedspi.aims.media.Media;

public class Store {
    private ArrayList<Media> itemsInStore = new ArrayList<Media>();

    public void addMedia(Media media) {
        itemsInStore.add(media);
        System.out.println(media.getTitle()+ " has been added to the store.");
    }

    public void removeMedia(Media media) {
        boolean found = false;
        for (Media item : itemsInStore) {
            if (item.equals(media)) {
                itemsInStore.remove(item);
                System.out.println(media.getTitle() + " has been removed from the store.");
                found = true;
                break;
            }
        }
        if (!found) System.out.println(media.getTitle() + " is not in the store.");
    }
    public ArrayList<Media> getItemsInStore() {return itemsInStore;}
}
```

9. Constructors of whole classes and parent classes

Update Use-Case Diagram



Update Class Diagram



10. Unique item in a list

Override the boolean equals() in Media and Track

```java
//<<<<<<< topic/override-equals-method
    public boolean equals(Object obj) {
        if (obj instanceof Track) {
            Track track = (Track) obj;
            if (this.titleTrack.equals(track.titleTrack) && this.length == track.length) {
                return true;
            }
        }
        return false;
    }
}
```

```java
    public boolean equals(Object obj) {
        if (obj instanceof Media) {
            Media media = (Media) obj;
            if (this.id == media.id) {
                return true;
            }
        }
        return false;
    }
```

11. Polymorphism with toString() method

```java
    Run | Debug
    public static void main(String[] args) {
        ArrayList<Media> mediae = new ArrayList<>();

        CompactDisc cd = new CompactDisc(id:1,title:"Soledad", category:"Ballad", cost:12.5f);
        DigitalVideoDisc dvd = new DigitalVideoDisc(id:3,title:"Final Fantasy X", category:"Fantasy", cost:222.22f );
        Book book = new Book(id:2,title:"Operating System Concepts", category:"ICT", cost:30f);

        // Add some media objects to the list
        mediae.add(cd);
        mediae.add(dvd);
        mediae.add(book);

        Collections.sort(mediae, Media.COMPARE_BY_TITLE_COST);

        for (Media media : mediae) {
            System.out.println(media.toString());
        }
    }
}
```

12. Sort media in the cart

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;

import java.util.Comparator;

public class MediaComparatorByCostTitle implements Comparator<Media> {
        public int compare(Media media1, Media media2) {
        return Comparator.comparing(Media::getCost)
                .thenComparing(Media::getTitle)
                .compare(media1, media2);
    }
}
```

```java
package lab04.Aimsproject.hust.soict.hedspi.aims.media;

import java.util.Comparator;

public class MediaComparatorByTitleCost implements Comparator<Media> {
        public int compare(Media media1, Media media2) {
        return Comparator.comparing(Media::getTitle)
                .thenComparing(Media::getCost)
                .compare(media1, media2);
    }
}
```

__Question__: Alternatively, to compare items in the cart, instead of using `Comparator`, we can use the `Comparable` interface and override the `compareTo()` method. You can refer to the Java docs to see the information of this interface.

Suppose we are taking this `Comparable` interface approach.

- **What class should implement the `Comparable` interface?**
  Media class (abstract class that contains compare objects) should implement.
- **In those classes, how should you implement the `compareTo()` method be to reflect the ordering that we want?**
  We should compare objects' attributes.
- **Can we have two ordering rules of the item (by title then cost and by cost then title) if we use this `Comparable` interface approach?**
  No, because compareTo() method only returns one int value.

- **Suppose the DVDs has a different ordering rule from the other media types, that is by title, then decreasing length, then cost. How would you modify your code to allow this?**

```
public class DVD extends Media implements Comparable {

    public int compareTo(DVD other) {

        if (!this.getTitle().equals(other.getTitle())) {

            return this.getTitle().compareTo(other.getTitle()); }

        else if (this instanceof DVD && other instanceof DVD) {

            return Integer.compare(((DVD) other).getLength(), ((DVD) this).getLength()); }

        else return Double.compare(this.getCost(), other.getCost());

    }
```

13. } Create a complete console application in the Aims class

```
Run | Debug
public static void main(String[] args) {
    CompactDisc test = new CompactDisc(id:1, title:"Test", category:"test", cost:1f);
    CompactDisc cd = new CompactDisc(id:1,title:"The Fellowship of the Ring",category:"Fantasy", cost:12.5f);
    DigitalVideoDisc dvd = new DigitalVideoDisc(id:2,title:"The Two Towers", category:"Fantasy", cost:222.22f );
    Book book = new Book(id:3,title:"The Return of the King", category:"Fantasy", cost:30f);
    store.addMedia(cd); store.addMedia(dvd); store.addMedia(book);
    store.addMedia(test);
    cart.addMedia(test);
    //for (Media item : store.getItemsInStore()) {
        //System.out.println(item.toString());
    //}
    Scanner aims = new Scanner(System.in);
    while (true) {
        showMenu();
        int option = aims.nextInt();
        aims.nextLine();
        switch (option) {
            case 1:
                viewStore(aims);
                break;
            case 2:
                updateStore(aims);
                break;
            case 3:
                seeCurrentCart(aims);
                break;
            case 0:
                System.out.println(x:"Exiting the application...");
                aims.close();
                return;
            default:
                System.out.println(x:"Invalid option. Please try again.");
                break;
        }
```

```
59\OneDrive\Documents\workspace\Java>  & 'C:\Program Files\Java\jdk-22\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp'
'C:\Users\84859\AppData\Roaming\Code\User\workspaceStorage\91994c1888168c06b2094f7c720bc3c2\redhat.java\jdt_ws\Java_ded981d0\bin' 'la
b04.Aimsproject.hust.soict.hedspi.aims.aims'
The Fellowship of the Ring has been added to the store.
The Two Towers has been added to the store.
The Return of the King has been added to the store.
Test has been added to the store.
Test has been added to the cart.
Hi Tran Manh Hung(20220068),
AIMS:
--------------------------------
1.View store
2.Update store
3.See current cart
0. Exit
--------------------------------
Please choose a number: 0-1-2-3
```